Cayley equivalence for short patterns

1200

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Abstract

We verify Conjecture 8.6 of Cerbai, Claesson, Ernst and Golab for all packed patterns of length at most five and for patterns of length six on at most two letters. In these ranges, total Cayley avoidance counts through length eleven already determine all fixed-alphabet avoidance counts. The proof combines exact finite distinguishing counts with the published word-Wilf equivalences of Jelínek and Mansour.

1 Statement and reduction

A packed word has a set of letters of the form $[k] = \{1, \dots, k\}$; the empty word has $k = 0$. An occurrence of a packed pattern is a subsequence with the same order relations and equalities. For a pattern p, let $A_p(n, k)$ count the avoiding packed words of length n on exactly $[k]$, let $A_p(n) = \sum_k A_p(n, k)$, and let $W_p(n, K)$ count avoiding words on $[K]$ without requiring every letter to occur.

Theorem 1. *Suppose p and q have the same positive length m. If $m \le 5$, or if $m = 6$ and both patterns use at most two distinct values, then the following conditions are equivalent:*

1. *$A_p(n) = A_q(n)$ for every $0 \le n \le 11$;*
2. *$A_p(n) = A_q(n)$ for every $n \ge 0$;*
3. *$A_p(n, k) = A_q(n, k)$ for every $n, k \ge 0$;*
4. *$W_p(n, K) = W_q(n, K)$ for every $n, K \ge 0$.*

Split a word on $[K]$ according to its set of used letters. Increasing relabelling preserves containment, so for each n,

$$W_p(n, K) = \sum_{k=0}^{K} \binom{K}{k} A_p(n, k).$$

This identity and binomial inversion give (3)⇔(4); summing gives (3)⇒(2), and (2)⇒(1) is immediate. We prove (1)⇒(4) by showing that distinct word-Wilf classes have distinct finite total signatures.

2 Class distinctions

Use the word-Wilf classification in [2, §§1.1, 1.4 and Appendix A]. The authors' table [3] lists representatives up to reversal and rank complement; both operations preserve fixed-alphabet avoidance. Expanding its rows covers exactly the 75 patterns of length four (16 classes), 541 of length five (116 classes), and 63 at-most-binary patterns of length six (18 classes), without overlap. Membership in the same row, after expansion, supplies the all-length implication.

There is one pattern of length one. At length two the classes represented by 11 and 12 have totals 6 and 4 at host length three. At length three the classes represented by $111, 112, 123$ have totals 66, 60, and 56 at host length four. At pattern lengths four and five, direct enumeration at host lengths eight and nine, respectively, separates all but the corresponding rows listed below. At pattern length six with at most two values, the vectors of totals at host lengths six through ten separate all but the last two rows. Each row of this table supplies a further host length that separates its pair.

m	Representatives (p, q)	n	$A_p(n)$	$A_q(n)$
4	1223, 1332	9	4,472,716	4,472,712
5	11212, 12121	10	98,890,988	98,890,980
5	11223, 11332	11	1,445,533,180	1,445,533,200
5	12123, 13232	10	93,909,363	93,909,357
5	12234, 12443	10	87,548,792	87,548,796
5	13442, 14233	10	87,410,614	87,410,626
6	112112, 121112	11	1,616,420,895	1,616,420,896
6	111212, 112121	11	1,616,424,886	1,616,424,885

Thus (1) forces membership in the same word-Wilf class, which gives (4) by the cited classification. This proves the stated part of Conjecture 8.6 [1, p. 295]. The original conjecture for arbitrary pattern length is outside the result.

3 Exact counts

For the two complete censuses, enumerate each host on its actual alphabet, retaining only words that use every value of that alphabet. At a leaf, standardize each increasing selection of positions and count each resulting pattern once for that host. The code checks the total population against the sum of the layers given by the surjection recurrence

$$F(n,k) = k\big(F(n-1,k) + F(n-1,k-1)\big), \qquad F(0,0) = 1.$$

Deleting the last letter either leaves all k letters or removes the only instance of that letter, giving the two terms. Subtracting containing hosts from the population gives avoidance.

For individual patterns, an alternative count starts at the pattern and inserts one letter at every step. Either the letter is already present, or it is a new rank and all old ranks at least as large are increased by one. In any larger containing word, deleting a position outside a fixed occurrence and standardizing gives a containing parent. Conversely either insertion preserves that occurrence. This proves coverage, and storing the words in sets handles repeated generation.

The large comparison 11223/11332 uses a streaming recognizer. Both patterns have the form (a, a, d, d, x), with $a < d < x$ for the first or $a < x < d$ for the second. While reading a prefix, maintain the minimum repeated value, the least $a < d$ for each realized prefix (a, a, d), and the possible final ranks following any (a, a, d, d). A new letter is tested as a final rank before updating shorter prefixes. The least a for each d gives the whole interval of possible middle ranks; for the other pattern, only the minimum completed d matters. If both patterns have occurred, their common remaining packed completions can be added by a recurrence on the remaining positions and unused letters. The streaming counts agree with insertion enumeration at lengths nine and ten. The accompanying archive includes source code, execution records, and a coverage checker.

AI contributions: GLM5.3F and GPT Astra. GLM5.3F contributed to the source research. GPT Astra assisted with the short-pattern classification, proof development, and independent verification.

References

[1] G. Cerbai, A. Claesson, D. C. Ernst and H. Golab, Pattern-avoiding Cayley permutations via combinatorial species, Australas. J. Combin. **94**(2) (2026), 267–299.

[2] V. Jelínek and T. Mansour, Wilf-equivalence on k-ary words, compositions, and parking functions, Electron. J. Combin. **16**(1) (2009), #R58.

[3] V. Jelínek and T. Mansour, Supplementary classification table for patterns of lengths four, five, and six.